LIBRARY MANAGEMENT SYSTEM

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Product Description

The Library Management System's goal is to create an efficient and effective way of managing various library systems, such as keeping track of borrowed books, basic information about patrons, a catalog of all the books available from different library branches, and much more. By using our management system, both employees and patrons will be able to have access to their library's extensive collection and accomplish various tasks online instead of having to go to the closest library branch. One key feature that our management system focuses on is the holds on library books. Some books tend to become more popular than others, resulting in high demand. However, due to the limited availability, not all patrons will have the ability to borrow and read the book. To resolve this issue, we plan on introducing a first come, first serve hold system to allow patrons to place a reservation on books that are currently being borrowed by their fellow patrons. This system would also keep track of patrons who have accessibility issues and would allow their books to be delivered instead of them having to pick up the books from their local library. Our database system will also store digital libraries such as audiobooks and research papers for those who need them. The database will also be designed in a way for the employees to view information regarding the patrons when they visit the library. Information such as overdue fines, near-due books, and preferred contact methods are displayed to easily help the employees assist their patrons.

Use Cases

1. **Use Case:** Limited availability of popular books

   Actor: Michelle (Librarian), Joshua (Library Patron)

   Description: The use case begins with Michelle who is a librarian at the Main Library. She's currently assisting patrons with borrowing books, one of which is this season's most popular book. Joshua, a frequent patron of the library, also wants to borrow the said book. Unfortunately for him, Michelle lets him know that there are no more copies available and that he will have to come back some other time when the books have been returned. However, Michelle lets Joshua know that there is a way for him to place a reservation on the book so that he would be informed when a copy has been returned to the library and will be available for him to borrow, to which Joshua agrees. Michelle uses the system to access Joshua's profile and puts a reminder that he has a reservation on the next book that will be returned. The system also has a way to prevent patrons from reserving too many books at a time. If the patron has more than ten book reservations, the system will ask the patron to either cancel one of their reservations or to reserve the book later after they have fewer than ten reservations.

2. **Use Case:** Patrons that needs better accessibility

   Actor: Sophia (Library Patron)

   Description: Sophia loves going to her local library and looks forward to going there every week to browse the newly arrived books. However, due to an injury, she was required to be homebound and would be required to use a wheelchair for the next few months until her feet recovers. Devastated that she will not be able to go to her local library, she uses the library's website and places an order for the books that she wishes to read and have them delivered to her house. To return the books, she will use the same packaging that the books came in and would ask someone to drop them off at the local post office. To make sure that all the books would get returned, each person is allowed to have ten books delivered to them, afterwards they would have to return a book if they wish to borrow more.

3. **Use Case:** Overdue books

   Actor: Jennifer (Librarian)

   Description: Jennifer is a librarian at the Main Library branch. She notices that there are books that have been borrowed for months and have yet to be returned. The library has a policy that states that books can be borrowed for two weeks and would then have to be returned to the library, or there would be a 10-cent fine per day that it was not returned. Jennifer now needs to find a way to find patrons who have overdue books and send them an email notification regarding the books and payment. She uses the library's database system to search for these patrons and sends them an invoice and reminders regarding their overdue books. The system also allows her to send a message to the patrons who have a hold on the books to kindly wait until they have more available copies of the book. Patrons who have not paid or returned their overdue books will be prohibited from borrowing another book until their fine is paid and the book is returned if it was not declared as lost.

4. **Use Case:** Catalog

   Actor: Julia (Library Manager)

   Description: Julia is currently the library manager at her branch. With the new year arriving, her branch receives a shipment of newly published books, waiting to be borrowed by their patrons. She needed a way to keep track of the newly acquired books, as well as a way for her patrons to know the books that the library currently has. She uses the library's database system to update its collection of books and once all the information regarding the books has been entered, she publicizes this catalog for the patrons to use. This allows the patrons to search through their catalog, as well as identify which books are available at certain branches. To make it easier for both the patrons and the librarians to figure out which book they currently have, the catalog has filters that can be used such as the author's last name, alphabetical search, ISBN, genre as well as the year of publication.

5. **Use Case:** Digital Library

   Actor: William (Library Patron)

   Description: William is a frequent library goer but has terrible vision even when he is wearing glasses. His favorite author has finally released a book after five years and is excited to borrow it from his local library. After borrowing the physical copy of the book, he finds that the print is too small for him to read and disappointingly returns the book to his local library. The local librarian informs him that he can use his library card and the library app to borrow the digital copy of the book instead, which has features such as changing the font sizes. Similar to physical copies being borrowed, each patron can borrow up to ten digital copies at a time, and they would have to return the borrowed digital copies if they wish to borrow more. To prevent copyright issues, copy-pasting, and screenshots would not be allowed inside the app.

Functional Database Requirements

1. General User
    1.1. Each user can only create one account using one email
    1.2. A general user is an unregistered user, registered user or library staff.
    1.3. A general user shall be able to browse for many items
2. Registered User
    2.1. A registered user is a general user
    2.2. A registered user can borrow many books or digital media
    2.3. Each user can create a hold for many books
    2.4. Every user shall have a unique library card number which shall be connected to their account
    2.5. Many users can place a hold for any books
    2.6. Users must renew their library card after many years
    2.7. A registered user shall be able to register to a mailing list
    2.8. A registered user shall have one validated address
    2.9. A registered user shall be able to choose a default payment method
    2.10. A registered user shall have one and only one account
    2.11. A registered user shall be able to modify their own account
3. Account
    3.1. An account shall belong to only one user
    3.2. An account shall have an email address and a password
4. Books
    4.1. A book can be checked out by many users

4.2. A book shall be placed on hold by many users

4.3. A book can be borrowed for many weeks

4.4. The library will have many copies of each book

5. Library Card

5.1. A library card shall have an expiration date

5.2. A library card shall have a unique number ID

6. Catalog

6.1. A catalog is a collection of books and digital library

6.2. A catalog shall contain authors and credits of books and digital library

7. Employees

7.1. An employee is a registered user

7.2. An employee is the only user that can add books

7.3. An employee is the only user that can delete books

7.4. An employee is the only user that can edit books

7.5. Every librarian can make book holds for a patron

7.6. An employee is the only user that can send emails to other users

7.7. Each employee shall have a specific role

7.8. Every library shall have a library manager which is also a employee

7.9. Each library employee shall be paid in an hourly basis

7.10. Each librarian is assigned to work in at least one library branch

7.11. Each librarian is able to send notifications to patrons regarding their fees

7.12. A library manager shall be able to modify any account

7.13. A library manager shall be able to delete an account

8. Multimedia Content

   8.1. A multimedia content shall have many authors

   8.2. A multimedia content is either a movie, music, research papers or eBook

9. eBooks

   9.1. An eBook shall be previewed by a general user

   9.2. An eBook shall be downloaded by a registered user

   9.3. An eBook shall be available in one or more languages

   9.4. An eBook shall be available to a registered user for many weeks

   9.5. An eBook shall have one or more authors

   9.6. An eBook shall have a publisher

10. Video

   10.1. A video shall be previewed by a general user

   10.2. A video shall be streamed by a registered user

   10.3. A video shall be able to be resumed as many times as needed

   10.4. A video shall be available to a registered user for many weeks

   10.5. A video shall have one or more credit

11. Music

   11.1. A music shall be previewed by a general user

   11.2. A music shall be streamed by a registered user

   11.3. A music shall be able to be resumed as many times as needed

   11.4. A music shall be available to a registered user for many weeks

   11.5. A music shall have one or more credit

12. Research Paper

12.1. A research paper shall be downloaded by a student or faculty staff

12.2. A research paper shall have one or more publishers

12.3. A research paper shall have one or more author

13. Universities

13.1. All students and faculty staff shall have an account

13.2. All students and faculty staff will use their ID to log in to their accounts

13.3. All students and faculty staff is a registered user

14. Payment Method

14.1. A payment method shall be paid using a bank account

14.2. A bank account payment method is a credit card or debit card

Non-functional Requirements

1. Security
    1.1. All pins and passwords and user information shall be hashed and encrypted
    1.2. Pins shall contain a 5 numbers and passwords shall contain at least 8 characters with a mixture of lower case, uppercase and numbers
    1.3. The database shall automatically be backed up at 11:59pm
    1.4. All users shall have the option for signing up for a two-factor authentication to secure their account
    1.5. Every unknown login location shall require a two-factor authentication sent to the user's email to verify their identity
2. Performance
    2.1. Queries for data shall be returned in at least 5ms
    2.2. More specific queries shall be returned in less than 10ms
    2.3. The database should be able to handle at least 100 requests at a time
3. Scalability
    3.1. The system shall be able to support at least 50 concurrent users at a time
    3.2. During standard network speed, the website and app shall be able to load in less than 10 seconds
4. Maintainability
    4.1. After a system failure, the database shall not take more than 3 hours to recover
5. Memory Usage
    5.1. Under extreme usage and stress, the database shall not consume more than 60% of the memory

6. Storage Capacity
   6.1. Each table shall be assigned 10mb of memory
   6.2. The maximum amount of memory that each table can take is 2gb
7. DBMS
   7.1. Separate tables will be used to store library patrons, employees, books, and articles

Entity Relationship Diagram